

Ramakrishna Saravanan

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EDUCATION

- **Northwestern University** Evanston, IL, USA
Master of Science in Artificial Intelligence Sep 2025 - Dec 2026 (*Expected*)
- **Indian Institute of Technology(IIT) Patna** Patna, India
B.Tech in Engineering Physics. (CPI: 8.03) Jul 2021 - May 2025

EXPERIENCE

- **Samsung Research Institute Bangalore (PRISM)** Jul '24 - Mar '25
R&D Intern, Visual Intelligence Team
 - Developed an on-device inference model to aid in thumbnail and keyframe extraction, enabling automatic selection of aesthetically pleasing and diverse frames to summarize a movie's content best.
 - Optimized the model pipeline to achieve an state of the art inference speed of 10% of video length and performance on par with time complex deep learning solutions.
- **SPARK Summer Research Internship** May '24 - Jul '24
Summer Research Intern, IIT Roorkee
 - Streamlined the defect identification process by integrating advanced imaging algorithms within the Tkinter application, resulting in a significant decrease in average processing time in manual data annotation from initial analysis to final reporting for researchers handling complex metal-ceramic samples.
 - Enhanced the performance of a U-Net model in automation of defect analysis by meticulously training and fine-tuning, achieving an impressive segmentation accuracy of 97.3% for over 24,600 pixel-like defects in high-resolution CT imaging data

PROJECTS

- **PINN Simulation and Interpretability** Github
Under Prof. Ajay D Thakur, IIT Patna Jul '24 - May '25
 - Implemented Physics-Informed Neural Networks (PINNs) for solving fluid flow through porous media (Brinkman–Forchheimer equation), validating results with numerical simulations.
 - Applied saliency mapping and weight matrix pruning to enhance interpretability in deep neural networks for breast cancer detection, demonstrating how sparsity and gradient-based attention shape model decisions.
- **Ethereum Phishing Scam Detection** Github
Under Prof. Joydeep Chandra, IIT Patna Jul '23 - May '24
 - Developed and trained a Graph Convolutional Network (GCN) using Pytorch Geometric on Ethereum blockchain transaction data, achieving 98% accuracy in detecting phishing accounts.
 - Implemented preprocessing pipelines including data cleaning, feature engineering of graph node attributes, and utilizing SMOTE to perform data augmentation to solve class imbalance.

SKILLS/FRAMEWORKS

Programming Languages: Python, SQL, R, MATLAB.

Relevant Skills: Computer Vision, Natural Language Processing, Generative AI, Retrieval-Augmented Generation (RAG), Vision-Language Modeling, Model Deployment, Feature Engineering, Data Modeling, Data Wrangling, Exploratory Data Analysis, Statistical Modeling, A/B Testing, Model Monitoring.

ML & Data Science: TensorFlow, PyTorch, Keras, Numpy, Scikit-Learn, Scikit-Image, OCR, PIL, OpenCV, Matplotlib, Tableau, PySpark, Streamlit, Power BI, Gen AI, Transformers, Vision Language Models, LLM, Git, FastAPI, Docker, Kubernetes, MLflow, Weights & Biases.